



الجامعة الإسلامية العالمية ماليزيا
INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA
بُونَيَّرِيْنَتِيْ اِيْنَالْاَرَابِيْجِيَا مُلْدِيْنِيَا
Garden of Knowledge and Virtue

TAWHIDIC EPISTEMOLOGY
LEADING THE WAY

UMMATIC EXCELLENCE
LEADING THE WORLD

KHALIFAH • AMĀNAH • IQRA' • RAHMATAN UL-ĀLAMĪN

IVY WITCH

NADA

NEW-ERA ASSISTANCE & DISTRIBUTION AGILITY

“When Employment Looks Healthy, Who is Still Struggling?”

DAX CHALLENGE 2026

MEET THE TEAM



Introduction

Employment data is updated every year, but poverty data is not, making it harder to know who is struggling between surveys.

NADA bridges this gap by using multiple indicators to track economic vulnerability across Pahang.

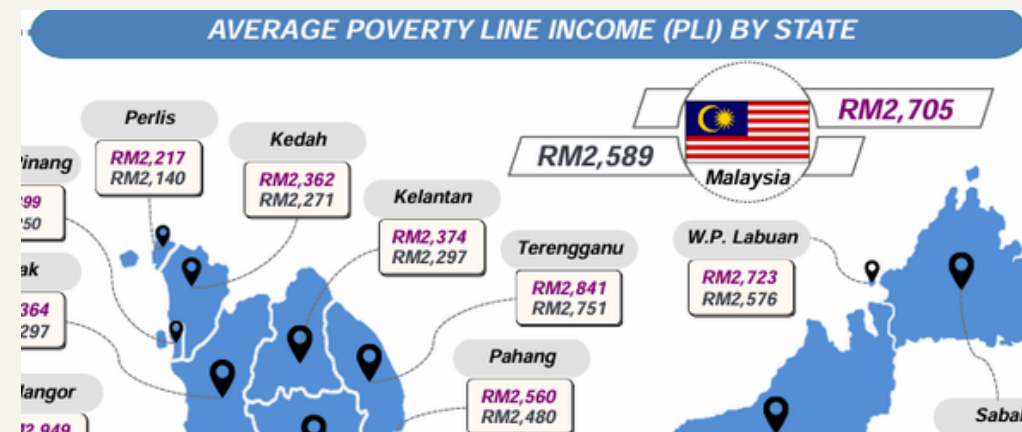


Problem Statement

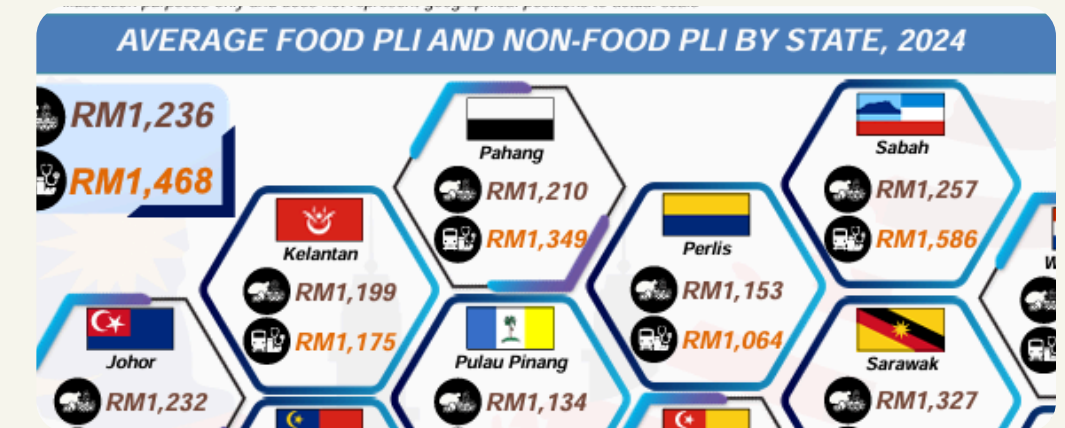
When Employment Doesn't Tell the Whole Story.



Why do Pahang's lowest-unemployment districts still suffer from some of the highest poverty rates?



How can the government distribute aid fairly when the household survey data we rely on is up to three years out of date?



How do we make Pahang's invisible "working poor" visible so that targeted support actually reaches the families who need it most?

The Double Blindspot



Economic vulnerability has multiple causes

Unemployment alone cannot capture factors such as inflation, dependency, amenities, labour participation, and agricultural exposure.

Young people lack clear, timely guidance

Gen Z may be employed but still face low income and rising living costs, making it difficult to know which actions can improve their financial future.

Relative and not absolute

The index measures relative risk between districts; it cannot calculate the exact, absolute number of poor households on its own



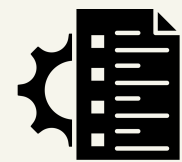
Data & Variables

11 Pahang Districts | 2016–2024 | Multiple Socioeconomic Indicators



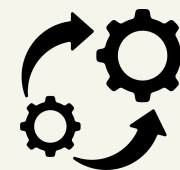
Ingestion

- OpenDOSM & MOE open-source data
- Multi-year temporal coverage
- Automated multi-source pulls



Standardisation

- Master district-year grid alignment
- Name crosswalking & harmonisation
- Zero primary key duplication check



Transformation

- Deprivation index inversion
- YoY macroeconomic calculation
- Demographic burden ratios



Modeling

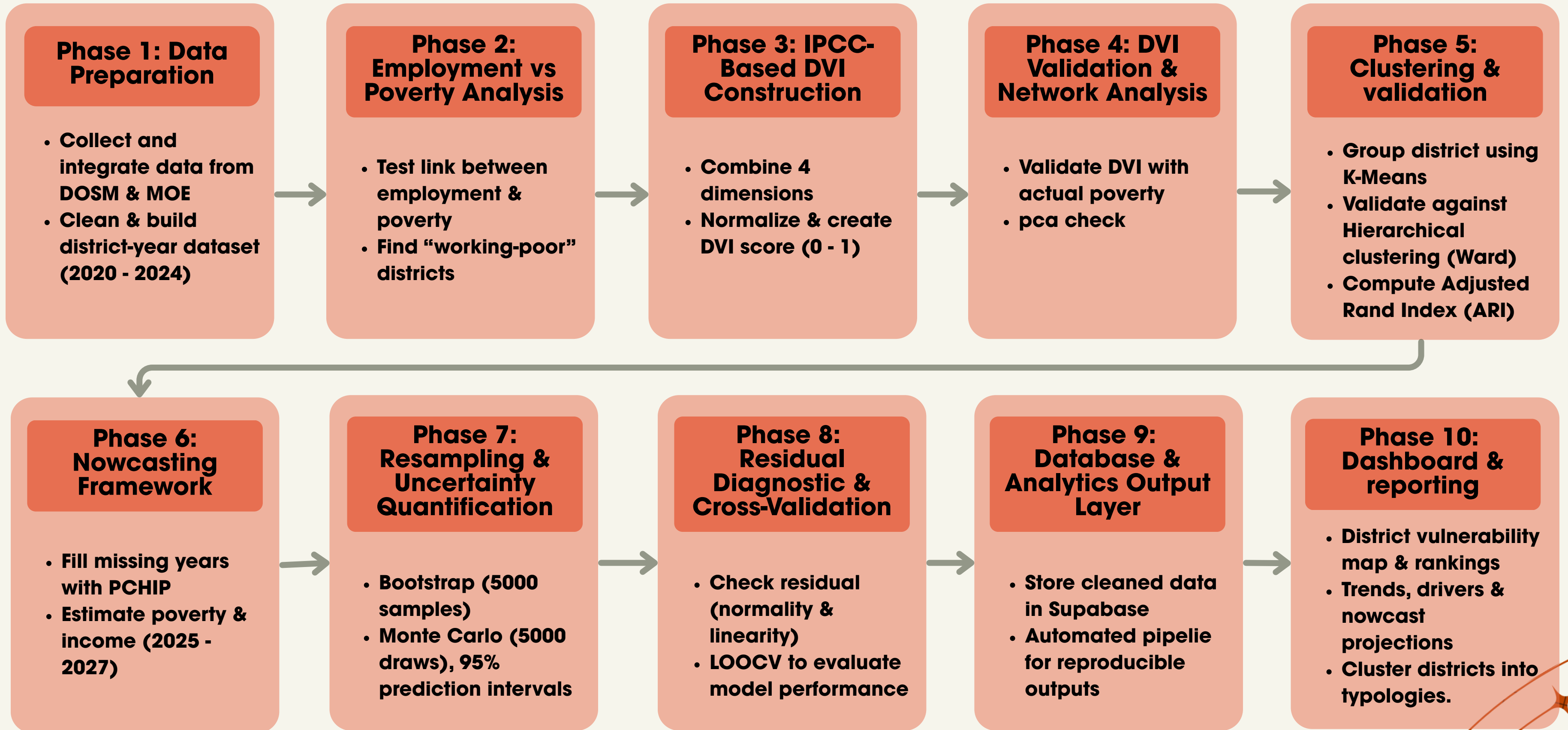
- PCHIP inter-survey gap interpolation
- Ridge regression & tree nowcasting
- Monte Carlo uncertainty sims



Deployment

- Supabase PostgreSQL cloud backend
- Power BI interactive executive UI
- Streamlit LLM tool-calling agent

Methodology





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DISTRICT VULNERABILITY
INTELLIGENCE — PAHANG



Overview & Labour
Mismatch

District Vulnerability
Index (DVI)

Nowcasting Engine

Definition

How to read

Youth Action
Guide

Data Sources: Department Of
Statistics Malaysia (DOSM)

Executive Overview & Labour Mismatch

Is low unemployment hiding poverty — and what does this mean for Pahang's young population? This page surfaces the gap between headline labour statistics and household-level economic health across Pahang's 11 districts

HIGHEST DISTRICT RISK

Maran

Score: 0.93

STATE AVERAGE
POVERTY RATE

6.22%

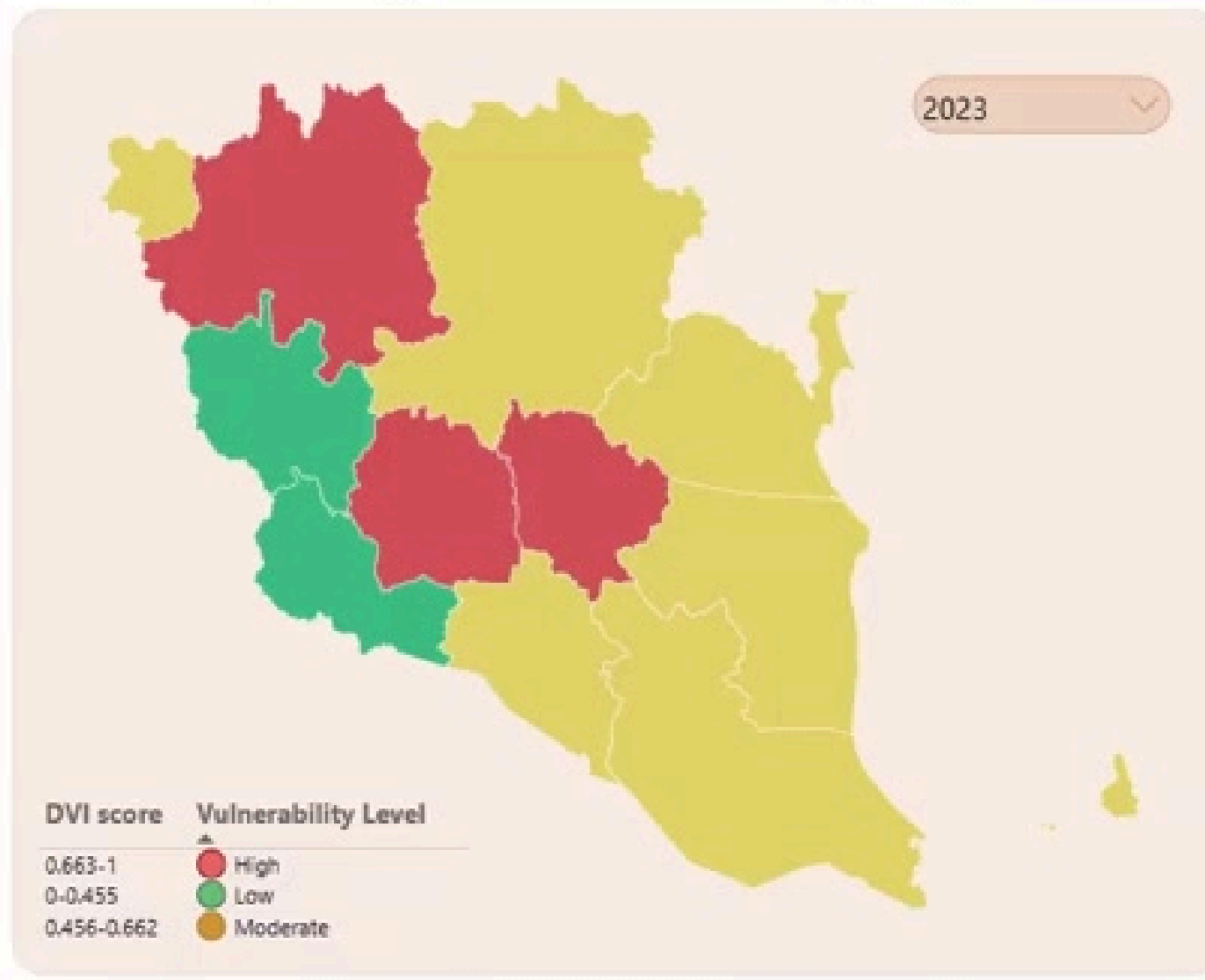
STATE AVERAGE LFPR

63.85%

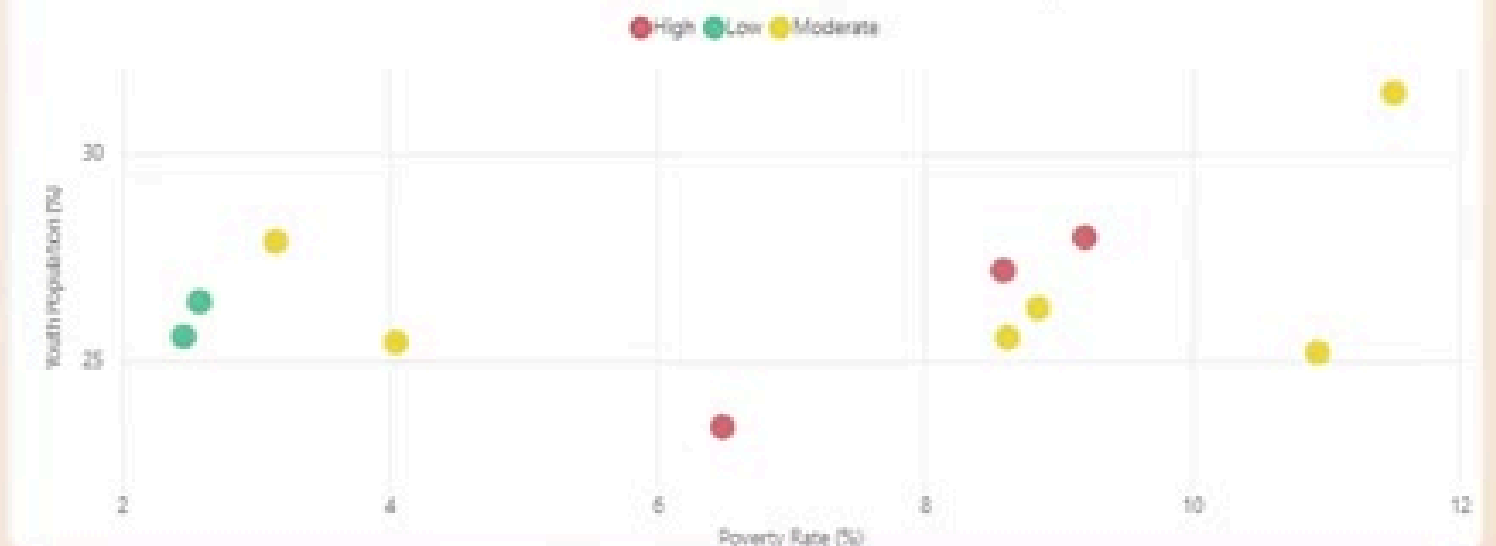
STATE AVERAGE YOUTH
POPULATION

25.78%

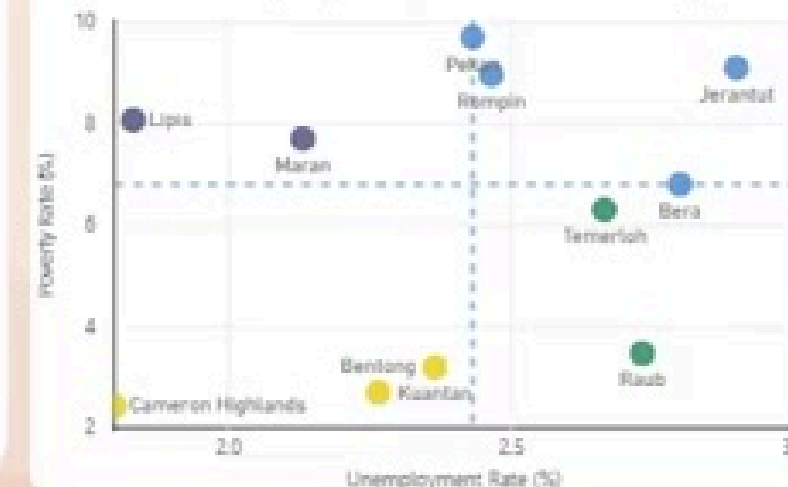
Pahang District Vulnerability Map



Youth Population vs. Poverty



Unemployment vs. Poverty



Unemployment vs. Poverty

- Low unemployment does not always mean low poverty.
- Lipis and Maran have low unemployment but higher poverty.

Youth Population vs. Poverty

- Districts with more young people do not always have higher poverty.
- Youth population alone cannot explain poverty.



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District Vulnerability Index (DVI)

HIGHEST DISTRICT RISK

Maran

Rank #1

DVI SCORE

0.99

Higher = Greater vulnerability

VULNERABILITY LEVEL

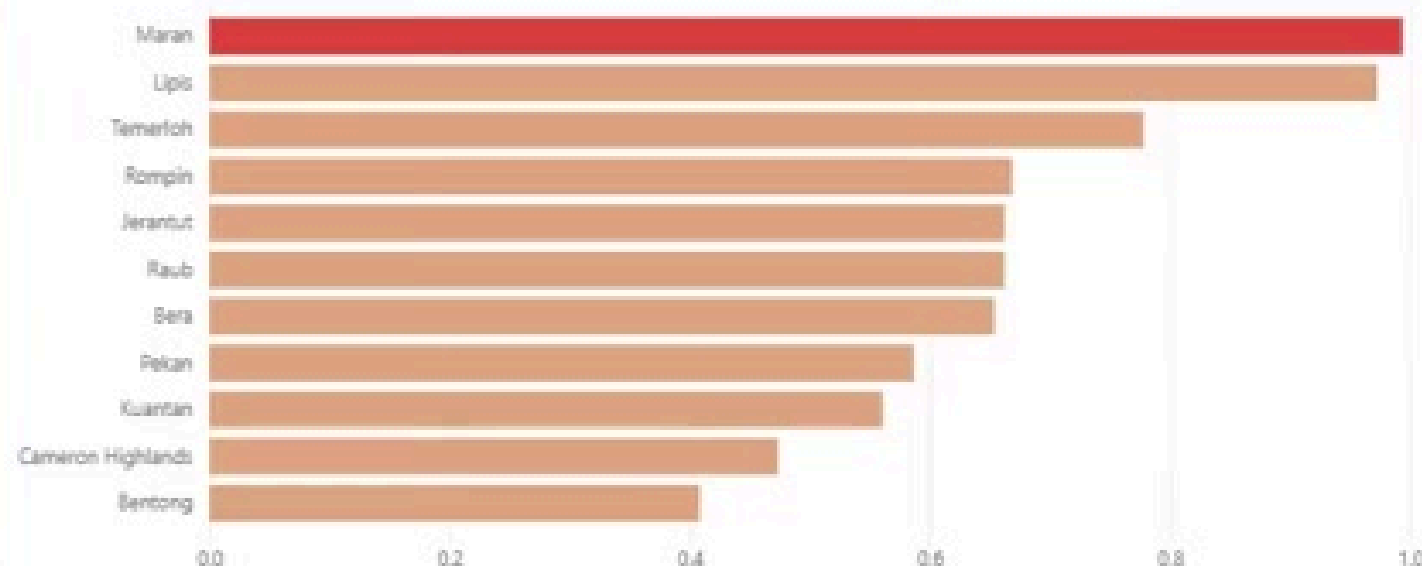
High

MAIN DRIVER

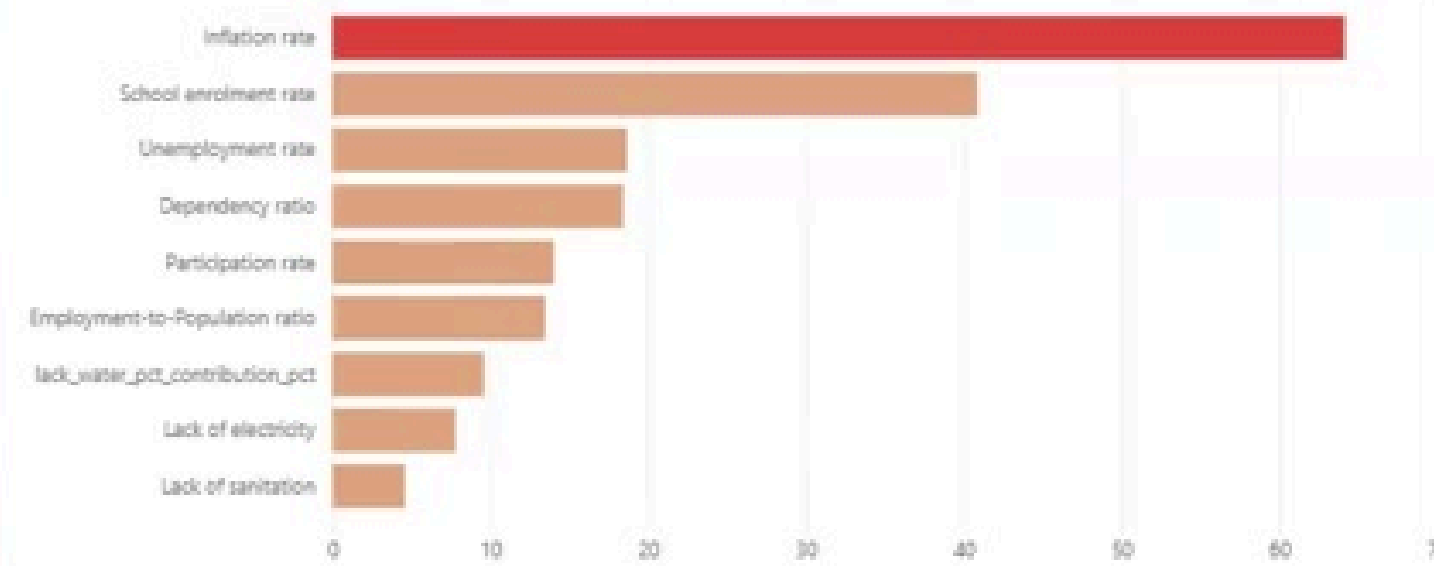
Inflation

Largest contributor to DVI

DVI Ranking Leaderboard



Indicator-DVI Correlation



Policy Action Summary

Year : 2022

District : All

District	Year	Vulnerability Level	Dominant Pillar	Dominant Indicator	Dominant Contribution
Bentong	2022	Low	Lack of Adaptive Capacity	School Enrolment	26.35
Bera	2022	Moderate	Exposure	Inflation	50.67
Cameron Highlands	2022	Moderate	Exposure	Inflation	69.96
Jerantut	2022	Moderate	Lack of Adaptive Capacity	School Enrolment	51.12
Kuantan	2022	Moderate	Exposure	Inflation	88.59
Lipis	2022	High	Exposure	Inflation	34.07
Maran	2022	High	Exposure	Inflation	100.00
Pekan	2022	Moderate	Exposure	Inflation	56.48
Raub	2022	Low	Lack of Adaptive Capacity	School Enrolment	73.04



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Economic Nowcasting & Uncertainty Simulation

Estimated present-day district median income and poverty rate for non-survey years, with clear communicated confidence ranges

DISTRICT : All

MODEL RELIABILITY : ● High — Income ● Moderate — Poverty

Pahang Est. Median Income (2026)

RM4,600.70

CI: RM4040.76 - RM4933.45

Pahang Est. Poverty Rate (2026)

7.62%

CI: 6.18% - 10.21%

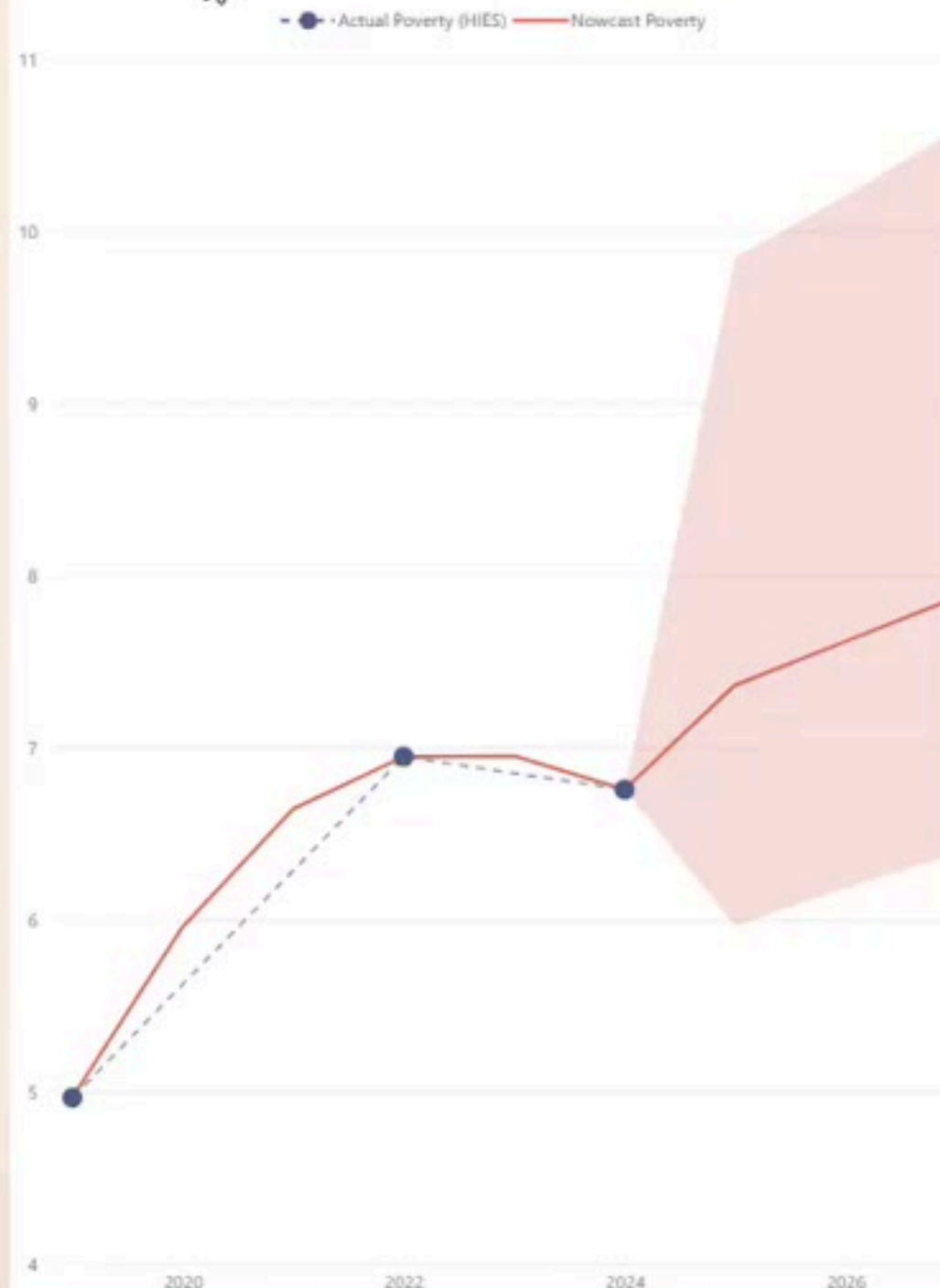
Model Reliability

Moderate-to-High

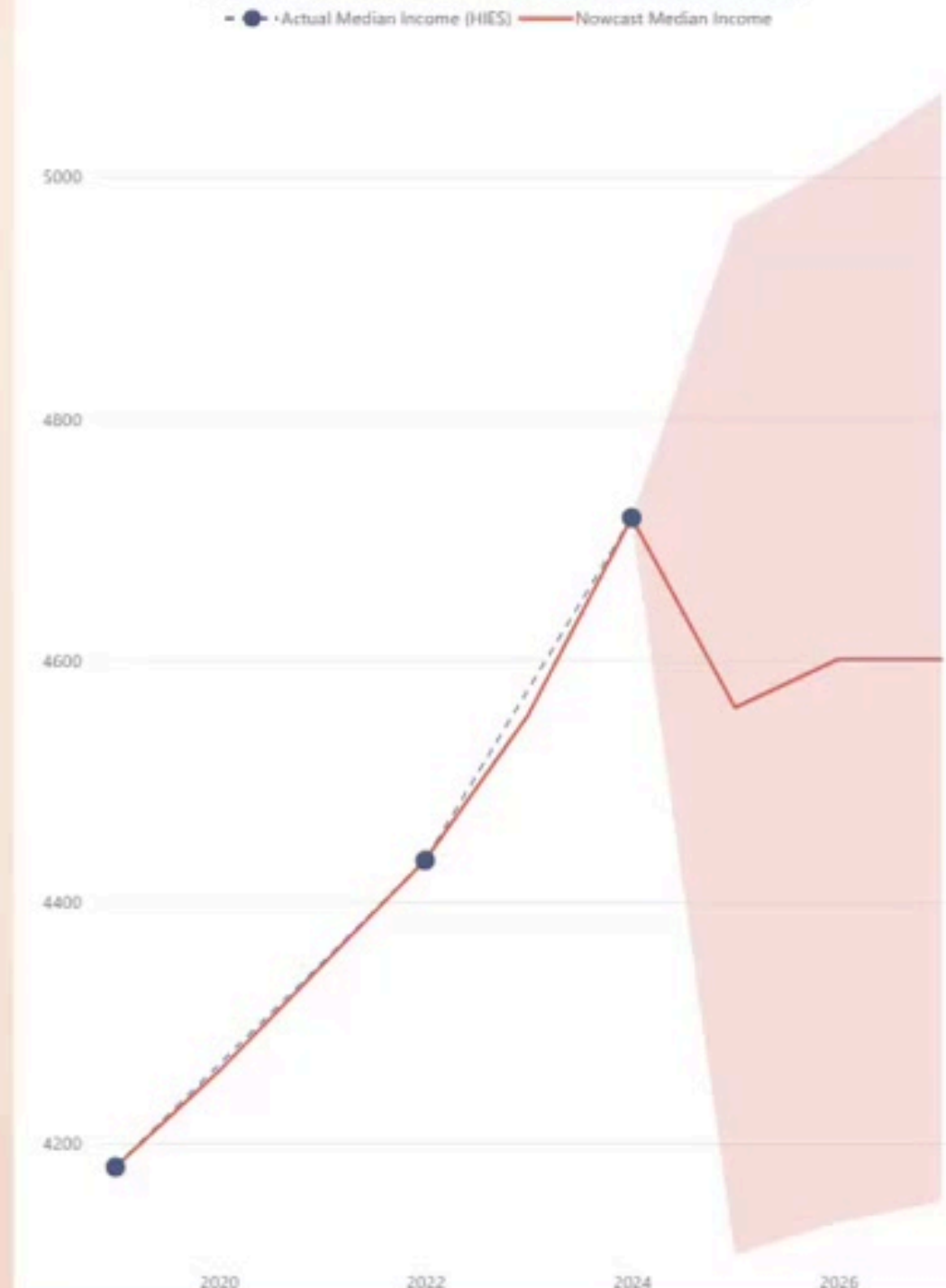
*Income: stronger predictive performance
Poverty: for identifying general trends*

Metric	Income	Poverty
LOOCV MAE	278.45	1.80
LOOCV RMSE	356.26	2.17
Validation MAPE	6.31	48.07
Validation R ²	0.66	0.53

Nowcasted Poverty Rate



Nowcasted Median Income



Strategic Recommendations for NGOs & Planners

NGOs and state planners must **look beyond headline unemployment rates and incorporate alternative indicators**, such as labour-market participation, to accurately identify districts experiencing hidden poverty.

NGOs should **utilize our District Vulnerability Index (DVI) dashboard** to evaluate **composite risk scores** and determine precisely which districts require priority assistance.

NGOs can **leverage our nowcasting model** to anticipate **future income and poverty trajectories** during inter-survey gap years, enabling **proactive and early preparation**.

THANK YOU